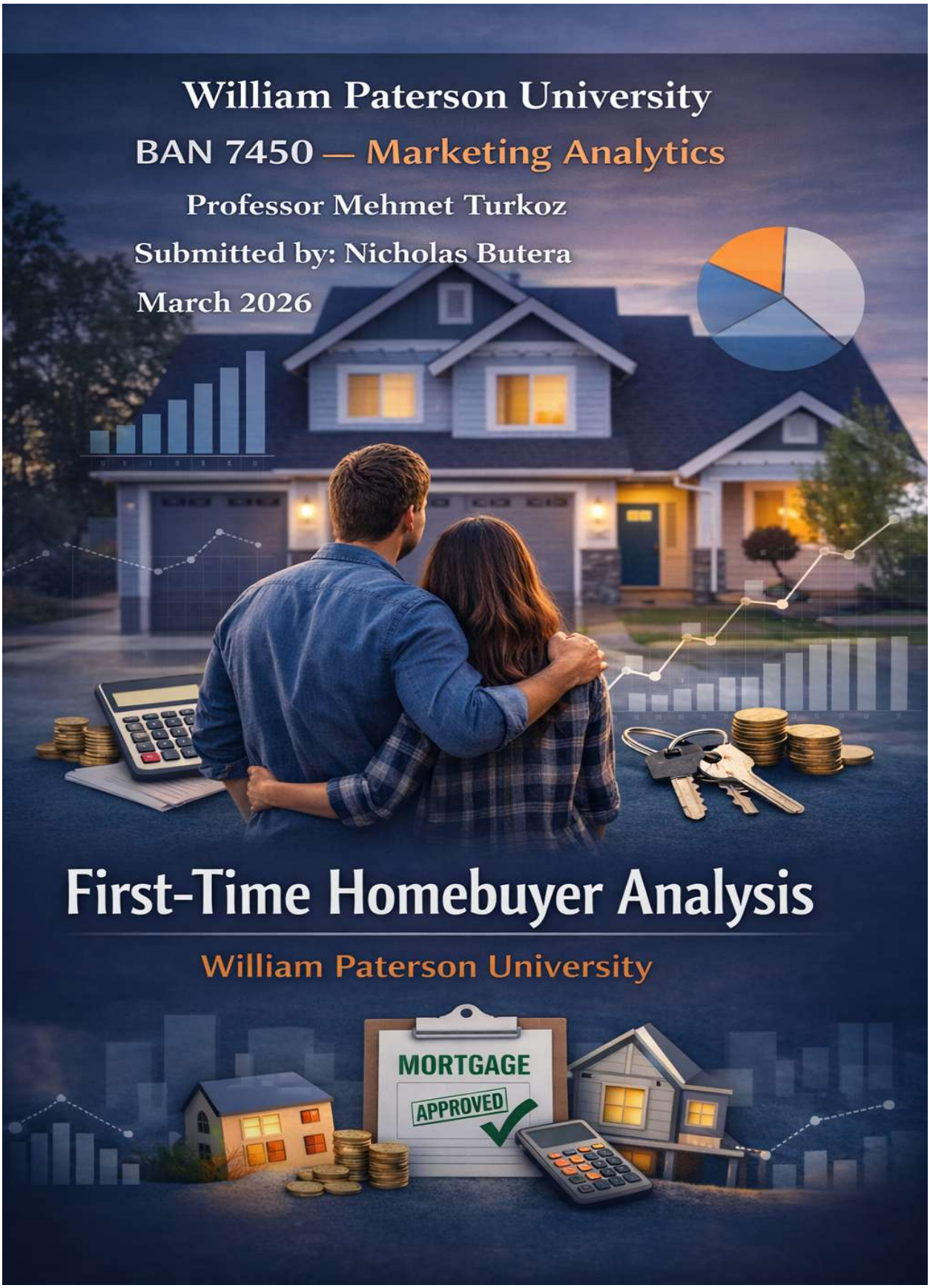
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BAN 7450 — Marketing Analytics
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First-Time Homebuyer Analysis
William Paterson University
MORTGAGE
APPROVED

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Introduction

Banks and mortgage lenders have been competing to attract and retain borrowers in the U.S. housing market for over 100 years. They do this by offering products that match customer needs and financial profiles. One important segment is first-time homebuyers, who often need more guidance, education, and product offerings compared to repeat buyers. Understanding what characteristics are most associated with being a first-time homebuyer can help a lender improve targeting, tailor offers, and allocate marketing resources more effectively.

A 2019 mortgage lending dataset containing borrower demographics, income and credit-related variables, loan terms, and property/area characteristics will be analyzed in this project. The goal of this is to identify which factors are most strongly related to whether a borrower is a first-time homebuyer.

To accomplish this, exploratory data analysis (EDA) will be used to compare first-time and non-first-time borrowers across key variables. Next, a logistic regression model is used to estimate the probability that an applicant is a first-time homebuyer based on selected borrower and loan features (e.g., age, income, loan-to-value ratio, debt-to-income ratio, loan amount, loan type). The results are summarized in business terms and used to propose marketing recommendations for how a lender could better reach and support first-time homebuyer segments.

Data & Methods

This project uses a 2019 mortgage lending dataset with 89,767 loan records and 56 variables. The dependent variable is BorrowerFirstTimeHomebuyer, which indicates whether the borrower is a first-time homebuyer (Yes/No). In this dataset, 31,971 borrowers are first-time homebuyers and 57,796 are non-first-time homebuyers.

The predictors used in the analysis include borrower age at application, total monthly income, loan-to-value ratio (LTV), debt-to-income ratio (DTI), loan amount, note rate, mortgage type, and loan purpose. The dataset was imported into RStudio and checked for missing values across the selected variables; no missing values were found in the modeling fields. Mortgage type and loan purpose were recoded into labeled categories to improve readability. To make results easier to interpret, income was scaled per $1,000 and loan amount was scaled per $10,000.

Exploratory Data Analysis

Exploratory data analysis was conducted to compare first-time and non-first-time borrowers across key variables. First-time homebuyers were slightly younger than non-first-time borrowers, with a median age of **41** compared to **43**. First-time borrowers also had lower typical income and loan size: the median monthly income was **$7,865** for first-time borrowers versus **$8,577** for non-first-time borrowers, and the median loan amount was **$211,000** versus **$224,000**, respectively.

Interest rates differed substantially between groups. The median note rate was **4.25%** for first-time borrowers compared to **3.75%** for non-first-time borrowers, suggesting differences in pricing and/or the loan programs used. Consistent with this, the mortgage type comparison shows that first-time borrowers have a higher share of non-conventional products (FHA, USDA,

and VA) than non-first-time borrowers. These patterns from the EDA informed the predictors used in the logistic regression model.

Table 1. Median values by first-time homebuyer status

First-time Homebuyer	n	age_median	income_median	ltv_median	dti_median	note_amt_median	rate_median
No	57,796	43	$8,577	79	31.19	$224,000	3.75
Yes	31,971	41	$7,865	80	32.74	$211,000	4.25

Table 2. Mean values by first-time homebuyer status

First-time Homebuyer	n	age_mean	income_mean	ltv_mean	dti_mean	note_amt_mean	rate_mean
No	57,796	44.71	$10,106.88	73.84	31.11	$247,221	3.80
Yes	31,971	43.76	$9,350.78	76.40	32.33	$238,351	4.27

Note: Medians are reported because income and loan amount distributions are right-skewed; means are included for comparison.

Figure 1. Age distribution by first-time homebuyer status

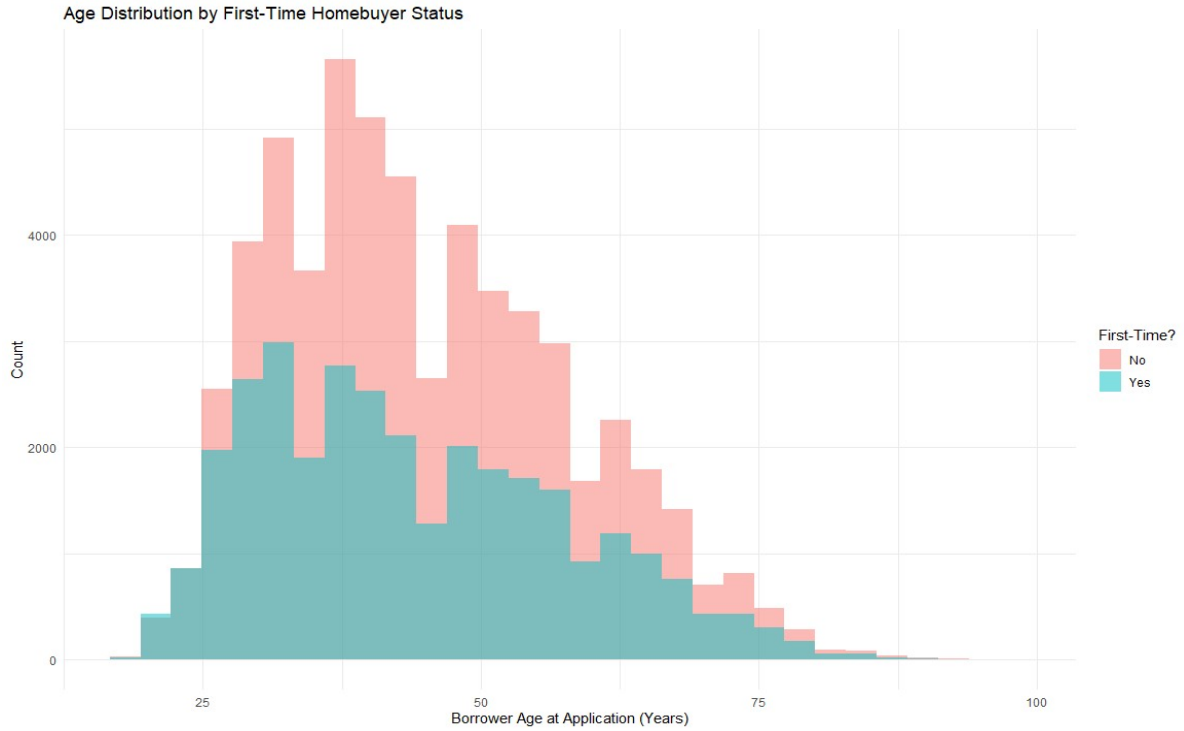


Figure 2. Monthly income distribution by first-time homebuyer status (log-scale)

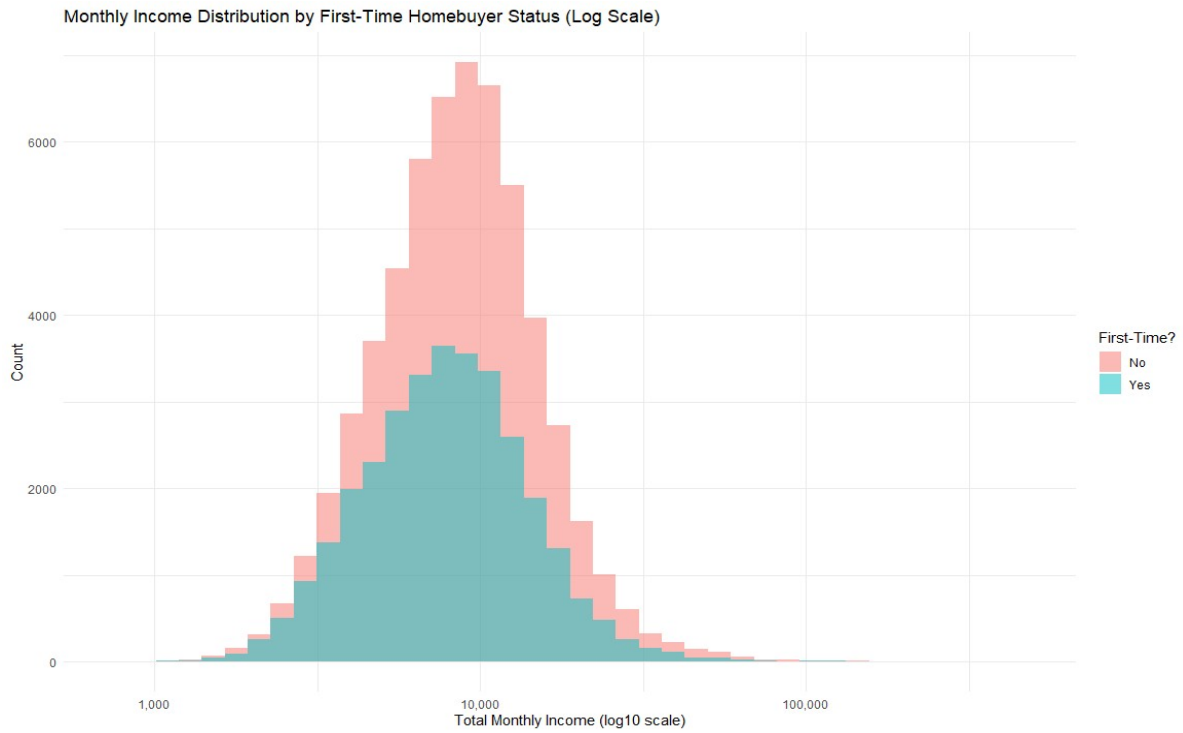
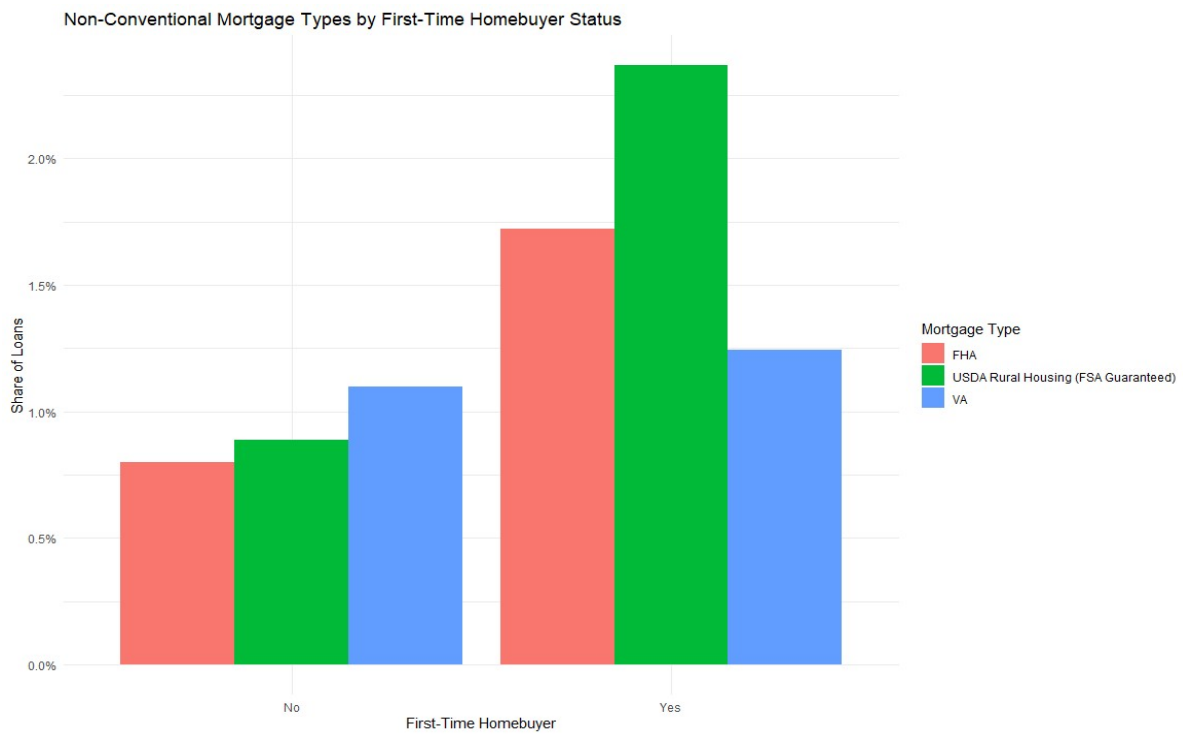


Figure 3. Non-conventional mortgage type mix by first-time homebuyer status



Logistic Regression

A logistic regression model was used to estimate the probability that a borrower is a first-time homebuyer based on borrower and loan characteristics. Logistic regression is appropriate because the dependent variable is binary and the model estimates how each predictor affects the odds of being a first-time homebuyer while holding other variables constant.

Using a 0.50 classification threshold, the model achieved an accuracy of **71.8%**. The model's **AUC was 0.795**, which indicates good ability to distinguish first-time from non-first-time borrowers across classification thresholds.

Odds ratios were used to interpret the model results. The strongest positive association with first-time status was **note rate percent (OR = 8.13)**. Mortgage type also contributed meaningfully: relative to conventional loans, the odds of being a first-time borrower were higher for **USDA (OR = 2.09)**, **FHA (OR = 1.69)**, and **VA (OR = 1.54)**. Loan purpose behaved as expected; compared to purchase loans, refinance loans were much less likely to be associated with first-time homebuyers (**Cash-out refi OR = 0.51; No-cash-out refi OR = 0.49**). Age and the financial ratio variables had smaller effects, but they were directionally consistent with the group comparisons observed in the EDA.

Table 3. Confusion matrix

	Actual No	Actual Yes
Predicted No	49069	16564
Predicted Yes	8727	15407

Table 4. Logistic Regression Odds Ratios (95% CI)

Predictor	OR	CI_low	CI_high
note_rate_percent	8.135	7.847	8.433
mortgage_type_labelUSDA Rural Housing (FSA Guaranteed)	2.093	1.837	2.385
mortgage_type_labelFHA	1.695	1.469	1.955
mortgage_type_labelVA	1.544	1.337	1.782
borrower1age_at_application_years	0.997	0.995	0.998
income_k	0.996	0.993	0.999
total_debt_expense_ratio_percent	0.995	0.993	0.997
loan_10k	0.995	0.993	0.996
ltv_ratio_percent	0.992	0.991	0.994
loan_purpose_labelCash-Out Refi	0.510	0.490	0.531
loan_purpose_labelNo-Cash Out Refi	0.494	0.473	0.515

Recommendations & Conclusion

The results suggest several practical actions for lenders. First, first-time marketing campaigns should be tailored to non-conventional programs, since FHA/USDA/VA loans are more strongly associated with first-time borrowers than conventional loans. Second, lenders should prioritize purchase-focused funnels for first-time outreach because refinance borrowers are unlikely to be first-time homebuyers. Third, because first-time borrowers are associated with higher rates in this dataset, lenders can test targeted incentives (such as closing-cost credits or rate buydowns) to improve competitiveness in first-time segments.

Overall, first-time homebuyers differ from non-first-time borrowers in age, income, loan size, interest rates, and mortgage program mix. The logistic regression model performed well (accuracy **71.8%**, AUC **0.795**) and identified note rate, mortgage type, and loan purpose as the strongest predictors of first-time status. These findings can be used to improve targeting and program-specific messaging for first-time homebuyer marketing.

Appendix

R Code:

```
## Marketing Analytics Project
# Step 1: Load + Quick Audit
install.packages(c("tidyverse", "janitor", "skimr", "ggplot2"))

library(tidyverse)
library(janitor)
library(skimr)
library(ggplot2)

# Load the CSV
df_raw <- read.csv(file.choose(), header = TRUE, stringsAsFactors = FALSE)
str(df_raw)
colnames(df_raw)
print(dim(df_raw))

# Standardize column names (lowercase, underscores)
df <- df_raw %>% janitor::clean_names()
colnames(df)

# Basic structure checks
cat("Rows, Cols:\n")
print(dim(df))

cat("\nColumn types (glimpse):\n")
glimpse(df)

cat("\nOverall missing values in entire dataset:\n")
print(sum(is.na(df)))

# Confirm target variable exists and how it's coded
# (Option A target: borrower_first_time_homebuyer)
cat('\nTarget value counts:\n"')
print(table(df$borrower_first_time_homebuyer, useNA = "ifany"))
```

```
# Convert target to factor labels (No/Yes) for modeling/report tables
df <- df %>%
  mutate(
    borrower_first_time_homebuyer = factor(
      borrower_first_time_homebuyer,
      levels = c(0, 1),
      labels = c("No", "Yes")
    )
  )

cat("\nTarget after labeling:\n")
print(table(df$borrower_first_time_homebuyer))

# Select the columns and audit them
model_cols <- c(
  "borrower_first_time_homebuyer",
  "borrower1age_at_application_years",
  "total_monthly_income_amount",
  "ltv_ratio_percent",
  "total_debt_expense_ratio_percent",
  "note_amount",
  "note_rate_percent",
  "mortgage_type",
  "loan_purpose_type"
)

model_df <- df %>% select(all_of(model_cols))
colnames(model_df)

cat("\nMissing values by model column\n")
print(colSums(is.na(model_df)))

cat("\nQuick numeric summaries:\n")
print(summary(model_df %>% select(
  borrower1age_at_application_years,
```

```
  total_monthly_income_amount,
  ltv_ratio_percent,
  total_debt_expense_ratio_percent,
  note_amount,
  note_rate_percent
)))

# Label coded categorical variables for readability in charts/tables
model_df <- model_df %>%
  mutate(
    loan_purpose_label = case_when(
      loan_purpose_type == 1 ~ "Purchase",
      loan_purpose_type == 2 ~ "No-Cash Out Refi",
      loan_purpose_type == 6 ~ "Cash-Out Refi",
      TRUE ~ "Other"
    ),
    mortgage_type_label = case_when(
      mortgage_type == 0 ~ "Conventional",
      mortgage_type == 1 ~ "FHA",
      mortgage_type == 2 ~ "VA",
      mortgage_type == 3 ~ "USDA Rural Housing (FSA Guaranteed)",
      TRUE ~ "Other"
    )
  )

cat("\nLoan purpose counts:\n")
print(table(model_df$loan_purpose_label, useNA = "ifany"))

cat("\nMortgage type counts:\n")
print(table(model_df$mortgage_type_label, useNA = "ifany"))

# Step 2: EDA (1 summary table + 2-3 plots)

install.packages(c("tidyverse","scales"))

library(tidyverse)
```

```
library(scales)

# A) Summary table (Report)
eda_table <- model_df %>%
  group_by(borrower_first_time_homebuyer) %>%
  summarise(
    n = n(),
    age_mean = mean(borrower1age_at_application_years),
    age_median = median(borrower1age_at_application_years),
    income_mean = mean(total_monthly_income_amount),
    income_median = median(total_monthly_income_amount),
    ltv_mean = mean(ltv_ratio_percent),
    ltv_median = median(ltv_ratio_percent),
    dti_mean = mean(total_debt_expense_ratio_percent),
    dti_median = median(total_debt_expense_ratio_percent),
    note_amt_mean = mean(note_amount),
    note_amt_median = median(note_amount),
    rate_mean = mean(note_rate_percent),
    rate_median = median(note_rate_percent),
    .groups = "drop"
  ) %>%
  mutate(
    income_mean = dollar(income_mean),
    income_median = dollar(income_median),
    note_amt_mean = dollar(note_amt_mean),
    note_amt_median = dollar(note_amt_median),
    age_mean = round(age_mean, 2),
    age_median = round(age_median, 2),
    ltv_mean = round(ltv_mean, 2),
    ltv_median = round(ltv_median, 2),
    dti_mean = round(dti_mean, 2),
    dti_median = round(dti_median, 2),
    rate_mean = round(rate_mean, 3),
    rate_median = round(rate_median, 3)
  )
```

```
print(eda_table, width = Inf)


# B) Plots (2-3)


# 1) Age Distribution

ggplot(model_df, aes(x = borrower1age_at_application_years, fill =
borrower_first_time_homebuyer)) +

  geom_histogram(bins = 30, position = "identity", alpha = 0.5) +

  labs(

    title = "Age Distribution by First-Time Homebuyer Status",

    x = "Borrower Age at Application (Years)",

    y = "Count",

    fill = "First-Time?"

  ) +

  theme_minimal()


ggsave("age_distribution.pdf", width = 8, height = 6)


# Income Distribution

ggplot(model_df, aes(x = total_monthly_income_amount, fill =
borrower_first_time_homebuyer)) +

  geom_histogram(bins = 40, position = "identity", alpha = 0.5) +

  scale_x_log10(labels = comma) +

  labs(

    title = "Monthly Income Distribution by First-Time Homebuyer Status (Log Scale)",

    x = "Total Monthly Income (log10 scale)",

    y = "Count",

    fill = "First-Time?"

  ) +

  theme_minimal()


ggsave("income_distribution.pdf", width = 8, height = 6)


# Mortgage type share (percent)

mortgage_share <- model_df %>%

  count(borrower_first_time_homebuyer, mortgage_type_label) %>%
```

```
  group_by(borrower_first_time_homebuyer) %>%
  mutate(pct = n / sum(n)) %>%
  ungroup()

print(mortgage_share)

ggplot(mortgage_share, aes(x = borrower_first_time_homebuyer, y = pct, fill =
mortgage_type_label)) +
  geom_col(position = "fill") +
  scale_y_continuous(labels = percent) +
  labs(
    title = "Mortgage Type Mix by First-Time Homebuyer Status",
    x = "First-Time Homebuyer",
    y = "Share of Loans",
    fill = "Mortage Type"
  ) +
  theme_minimal()

mortgage_share %>%
  filter(mortgage_type_label != "Conventional") %>%
  ggplot(aes(x = borrower_first_time_homebuyer, y = pct, fill = mortgage_type_label))
+
  geom_col(position = "dodge") +
  scale_y_continuous(labels = scales::percent) +
  labs(
    title = "Non-Conventional Mortgage Types by First-Time Homebuyer Status",
    x = "First-Time Homebuyer",
    y = "Share of Loans",
    fill = "Mortgage Type"
  ) +
  theme_minimal()

ggsave("mortgage_type_fthb.pdf", width = 8, height = 6)

# Step 3: Logistic Regression
library(tidyverse)
```

```
# 1) Make sure categorical variables are factors
model_df <- model_df %>%
  mutate(
    mortgage_type_label = factor(mortgage_type_label),
    loan_purpose_label = factor(loan_purpose_label)
  )

# Set reference categories so interpretation is clean
model_df <- model_df %>%
  mutate(
    mortgage_type_label = relevel(mortgage_type_label, ref = "Conventional"),
    loan_purpose_label = relevel(loan_purpose_label, ref = "Purchase")
  )

# 2) Fit logistic regression model
logit_fit <- glm(
  borrower_first_time_homebuyer ~
    borrower1age_at_application_years +
    total_monthly_income_amount +
    ltv_ratio_percent +
    total_debt_expense_ratio_percent +
    note_amount +
    note_rate_percent +
    mortgage_type_label +
    loan_purpose_label,
  data = model_df,
  family = binomial()
)

summary(logit_fit)

# 3) Odds ratios + 95% CI table (much easier to interpret than raw log-odds)
or_table <- tibble(
  term = names(coef(logit_fit)),
  estimate = coef(logit_fit),
```

```
  OR = exp(coef(logit_fit))
) %>%
  # Add Wald 95% CI
  mutate(
    se = sqrt(diag(vcov(logit_fit))),
    CI_low = exp(estimate - 1.96 * se),
    CI_high = exp(estimate + 1.96 * se)
  ) %>%
  select(term, OR, CI_low, CI_high) %>%
  arrange(desc(OR))

print(or_table, n = 50)

# The ORs for note amount and total monthly income are measured in 1 dollar units, so
a "1 unit" change is tiny
# Rescale the money variables. For income, interpret them per $1,000, and for loan
amount per $10,000.
names(model_df)

model_df2 <- model_df %>%
  mutate(
    income_k = total_monthly_income_amount / 1000,
    loan_10k = note_amount / 10000
  )

names(model_df2)

logit_fit2 <- glm(
  borrower_first_time_homebuyer ~
    borrower1age_at_application_years +
    income_k +
    ltv_ratio_percent +
    total_debt_expense_ratio_percent +
    loan_10k +
    note_rate_percent +
    mortgage_type_label +
```

```
    loan_purpose_label,
  data = model_df2,
  family = binomial()
)

# Odds ratios for the rescaled model
or_table2 <- tibble(
  term = names(coef(logit_fit2)),
  estimate = coef(logit_fit2),
  OR = exp(coef(logit_fit2)),
  se = sqrt(diag(vcov(logit_fit2)))
) %>%
  mutate(
    CI_low = exp(estimate - 1.96 * se),
    CI_high = exp(estimate + 1.96 * se)
  ) %>%
  select(term, OR, CI_low, CI_high) %>%

print(or_table2, n = 50)

or_table2_sorted <- or_table2 %>% arrange(desc(OR))
print(or_table2_sorted, n = 50)

# 4) Simple model performance (train-set accuracy + confusion matrix)
model_df2$pred_prob <- predict(logit_fit2, type = "response")

# Choose a threshold; 0.5 is standard
threshold <- 0.5
model_df2$pred_class <- ifelse(model_df2$pred_prob >= threshold, "Yes", "No") %>%
  factor(levels = c("No", "Yes"))

actual <- model_df2$borrower_first_time_homebuyer

conf_mat <- table(Predicted = model_df2$pred_class, Actual = actual)
print(conf_mat)
```

```
accuracy <- sum(diag(conf_mat)) / sum(conf_mat)
cat("\nAccuracy:", round(accuracy, 4), "\n")

# 5) AUC
install.packages("pROC")
library(pROC)

roc_obj <- roc(actual, model_df2$pred_prob, levels = c("No", "Yes"), direction = "<")
cat("AUC:", round(as.numeric(auc(roc_obj)), 4), "\n")

# Help put tables into Word
# Medians table
eda_medians <- eda_table %>%
  select(
    borrower_first_time_homebuyer, n,
    age_median, income_median, ltv_median, dti_median,
    note_amt_median, rate_median
  )

write.csv(eda_medians, "Table_1_Medians_by_FirstTimeStatus.csv", row.names = FALSE)

# Means table
eda_means <- eda_table %>%
  select(
    borrower_first_time_homebuyer, n,
    age_mean, income_mean, ltv_mean, dti_mean,
    note_amt_mean, rate_mean
  )

write.csv(eda_means, "Table_2_Means_by_FirstTimeStatus.csv", row.names = FALSE)

# OR table
or_report <- or_table2_sorted %>%
  filter(term != "(Intercept)") %>%
  mutate(
    OR = round(OR, 3),
```

```
    CI_low = round(CI_low, 3),
    CI_high = round(CI_high, 3)
  )

write.csv(or_report, "Table_OR_OddsRatios.csv", row.names = FALSE)
```